

FORMULA SHEET FOR STATISTICAL INFERENCE

INFERENCE FOR SINGLE POPULATION

Test Statistics for Mean

Test Statistic $z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$ and $t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$ with $d.f = n-1$

Critical Regions and P-Values for Z-test and T-test

	CR for z-test	CR for t-test	P-values for z-test	P-values for t-test
If H_1 contains $\neq$ sign	$ z \geq z_{\frac{\alpha}{2}}$	$ t \geq t_{\frac{\alpha}{2}(d.f)}$	$P = 2[1 - \Phi(z_0)]$	$P = 2[\Phi(t_0)]$
If H_1 contains $<$ sign	$z < -z_{\alpha}$	$t < -t_{\alpha(d.f)}$	$P = \Phi(z_0)$	$P = 1 - \Phi(t_0)$
If H_1 contains $>$ sign	$z > z_{\alpha}$	$t > t_{\alpha(d.f)}$	$P = 1 - \Phi(z_0)$	$P = \Phi(t_0)$

Confidence Intervals for Z-test and T-test

	Confidence Intervals for z-test	Confidence Intervals for t-test
If H_1 contains $\neq$ sign	$\bar{x} - Z_{\frac{\alpha}{2}} \cdot \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{x} + Z_{\frac{\alpha}{2}} \cdot \frac{\sigma}{\sqrt{n}}$	$\bar{x} - t_{\frac{\alpha}{2}(d.f)} \cdot \frac{S}{\sqrt{n}} \leq \mu \leq \bar{x} + t_{\frac{\alpha}{2}(d.f)} \cdot \frac{S}{\sqrt{n}}$
If H_1 contains $<$ sign	$\mu \leq \bar{x} + Z_{\alpha} \cdot \frac{\sigma}{\sqrt{n}}$	$\mu \leq \bar{x} + t_{\alpha(d.f)} \cdot \frac{S}{\sqrt{n}}$
If H_1 contains $>$ sign	$\mu \geq \bar{x} - Z_{\alpha} \cdot \frac{\sigma}{\sqrt{n}}$	$\mu \geq \bar{x} - t_{\alpha(d.f)} \cdot \frac{S}{\sqrt{n}}$

Test Statistics for Variance

Test Statistic $\chi^2 = \frac{(n-1)S^2}{\sigma^2}$ with $d.f = n-1$

Critical Regions and P-Values for Chi-square-test

	CR for chi-square test	P-values for chi-square-test
If H_1 contains $\neq$ sign	$\chi^2 > \chi_{\frac{\alpha}{2}, n-1}^2$ or $\chi^2 < \chi_{1-\frac{\alpha}{2}, n-1}^2$	search in table
If H_1 contains $<$ sign	$\chi^2 < \chi_{1-\alpha, n-1}^2$	search in table
If H_1 contains $>$ sign	$\chi^2 > \chi_{\alpha, n-1}^2$	search in table

Confidence Intervals for Variance

	Confidence Intervals for variance
If H_1 contains $\neq$ sign	$\frac{(n-1)S^2}{\chi_{\frac{\alpha}{2}, (n-1)}^2} \leq \sigma^2 \leq \frac{(n-1)S^2}{\chi_{(1-\frac{\alpha}{2}), (n-1)}^2}$
If H_1 contains $<$ sign	$\sigma^2 \leq \frac{(n-1)S^2}{\chi_{(1-\alpha), (n-1)}^2}$
If H_1 contains $>$ sign	$\sigma^2 \geq \frac{(n-1)S^2}{\chi_{\alpha, (n-1)}^2}$

Test Statistics for Proportion

Test Statistic $Z = \frac{\hat{p} - p_0}{\sqrt{p_0 q_0 / n}}$

Critical Regions and P-Values for Z-test

Note: CR and P-values are same as for mean

Confidence Intervals for Proportion

	Confidence Intervals for Proportion
If H_1 contains $\neq$ sign	$\hat{p} - Z_{\frac{\alpha}{2}} \sqrt{\frac{\hat{p}\hat{q}}{n}} < P < \hat{p} + Z_{\frac{\alpha}{2}} \sqrt{\frac{\hat{p}\hat{q}}{n}}$
If H_1 contains $<$ sign	$P \leq \hat{p} + Z_{\alpha} \sqrt{\frac{\hat{p}\hat{q}}{n}}$
If H_1 contains $>$ sign	$P \geq \hat{p} - Z_{\alpha} \sqrt{\frac{\hat{p}\hat{q}}{n}}$

INFERENCE FOR TWO POPULATIONS

Test Statistics for Difference of Means

$$\text{Test Statistic } Z = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \quad (\text{when } \sigma_1^2 \text{ and } \sigma_2^2 \text{ are known})$$

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} \quad \text{with } d.f = \frac{(s_1^2/n_1 + s_2^2/n_2)^2}{(s_1^2/n_1)^2/(n_1-1) + (s_2^2/n_2)^2/(n_2-1)} \quad (\text{when } \sigma_1^2 \text{ and } \sigma_2^2 \text{ are unknown but } \sigma_1^2 \neq \sigma_2^2)$$

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \quad \text{with } d.f = n_1 + n_2 - 2 \quad \text{where } S_p = \sqrt{\frac{(n_1-1)S_1^2 + (n_2-1)S_2^2}{n_1 + n_2 - 2}} \quad (\text{when } \sigma_1^2 \text{ and } \sigma_2^2 \text{ are unknown but } \sigma_1^2 = \sigma_2^2)$$

Critical Regions, P-Values and CI for Z-test and T-test

Note: CR and P-values are same as for single mean, for confidence intervals use the method of constructing CI from test-statistic as discussed in class.

Test Statistics for Paired Observations

$$\text{Test Statistic } t = \frac{\bar{d} - \mu_d}{S_d / \sqrt{n}} \quad \text{with } d.f = n - 1 \quad \text{Note that } \mu_d = \mu_1 - \mu_2$$

Note: CR and P-values are same as for single mean, for confidence intervals use the method of constructing CI from test-statistic as discussed in class.

Test Statistics for Variances

$$\text{Test Statistic } F = \frac{S_1^2}{S_2^2} \quad \text{if } S_1^2 > S_2^2 \quad \text{with } d.f = (n_1 - 1, n_2 - 1)$$

$$F = \frac{S_2^2}{S_1^2} \quad \text{if } S_2^2 > S_1^2 \quad \text{with } d.f = (n_2 - 1, n_1 - 1)$$

Critical Regions and CI for F-test

Critical region is given by $F \geq F_{\alpha(d.f)}$

$$\text{Confidence Interval for } \frac{\sigma_1^2}{\sigma_2^2} \text{ is given by } \frac{S_1^2}{S_2^2} F_{1-\alpha/2(d.f)} \leq \frac{\sigma_1^2}{\sigma_2^2} \leq \frac{S_1^2}{S_2^2} F_{\alpha/2(d.f)}$$

Test Statistics for Difference of Proportions

$$\text{Test Statistic } Z = \frac{(\hat{p}_1 - \hat{p}_2) - (P_1 - P_2)}{\sqrt{\frac{\hat{p}_1 \hat{q}_1}{n_1} + \frac{\hat{p}_2 \hat{q}_2}{n_2}}}$$

Critical Regions, P-Values and CI for Z-test

Note: CR and P-values are same as for single mean, for confidence intervals use the method of constructing CI from test-statistic as discussed in class.

FORMULAS FOR ANOVA

Table 12.3 ANOVA Table

Source of Variation	Degrees of Freedom	Sum of Squares	Mean Square	Value of the Test Statistic
Between	$k - 1$	SSB	MSB	$F = \frac{MSB}{MSW}$
Within	$n - k$	SSW	MSW	
Total	$n - 1$	SST		

$$\text{where } SSB = \left(\frac{T_1^2}{n_1} + \frac{T_2^2}{n_2} + \frac{T_3^2}{n_3} + \dots \right) - \frac{(\sum x)^2}{n}, \quad SSW = \sum x^2 - \left(\frac{T_1^2}{n_1} + \frac{T_2^2}{n_2} + \frac{T_3^2}{n_3} + \dots \right)$$

$$\text{also } MSB = \frac{SSB}{k-1} \quad \text{and} \quad MSW = \frac{SSW}{n-k}$$

SIMPLE LINEAR REGRESSION AND CORRELATION

$$r = \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{[n\sum x^2 - (\sum x)^2][n\sum y^2 - (\sum y)^2]}}$$

$$\text{For } Y = \alpha + \beta X + \varepsilon, \quad \hat{\beta} = \frac{n(\sum xy) - (\sum x)(\sum y)}{n(\sum x^2) - (\sum x)^2} \quad \text{and} \quad \hat{\alpha} = \bar{Y} - \hat{\beta}\bar{X}$$

$$R^2 = \frac{SSR}{SST} = \frac{\sum (\hat{Y} - \bar{Y})^2}{\sum (Y - \bar{Y})^2} \quad \text{or} \quad R^2 = 1 - \frac{SSE}{SST} = 1 - \frac{\sum (Y - \hat{Y})^2}{\sum (Y - \bar{Y})^2}$$

TESTING OF INTERCEPT α

$$\text{Test Statistic } t = \frac{\hat{\alpha} - \alpha}{S.E(\hat{\alpha})} \quad \text{where } S.E(\hat{\alpha}) = \sqrt{\frac{\sigma_e^2 \sum X^2}{n \cdot \sum (X - \bar{X})^2}} \quad \text{also } \sigma_e^2 = \frac{\sum e^2}{n-2} \quad CR: |t_{cal}| \geq t_{\frac{\alpha}{2}(n-2)}$$

The $(1 - \alpha)$ 100% for confidence interval for $\hat{\alpha}$ is

$$\hat{\alpha} - t_{\frac{\alpha}{2}(n-2)} \cdot S.E(\hat{\alpha}) \leq \alpha \leq \hat{\alpha} + t_{\frac{\alpha}{2}(n-2)} \cdot S.E(\hat{\alpha})$$

TESTING OF INTERCEPT β

$$\text{Test Statistic } t = \frac{\hat{\beta} - \beta}{S.E(\hat{\beta})} \quad \text{where } S.E(\hat{\beta}) = \sqrt{\frac{\sigma_e^2}{\sum (X - \bar{X})^2}} \quad \text{also } \sigma_e^2 = \frac{\sum e^2}{n-2} \quad \text{and } CR: |t_{cal}| \geq t_{\frac{\alpha}{2}(n-2)}$$

The $(1 - \alpha)$ 100% for confidence interval for β is

$$\hat{\beta} - t_{\frac{\alpha}{2}(n-2)} \cdot S.E(\hat{\beta}) \leq \beta \leq \hat{\beta} + t_{\frac{\alpha}{2}(n-2)} \cdot S.E(\hat{\beta})$$

FORMULAS FOR MLRM MATRICES APPROACH

For MLRM $Y = XB + E$

1 – Design Matrix

$$X^t X = \begin{bmatrix} n & \sum x_1 & \sum x_2 & \cdot & \cdot & \sum x_k \\ \sum x_1 & \sum x_1^2 & \sum x_1 x_2 & \cdot & \cdot & \sum x_1 x_k \\ \sum x_2 & \sum x_2 x_1 & \sum x_2^2 & \cdot & \cdot & \sum x_2 x_k \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \sum x_k & \sum x_k x_1 & \sum x_k x_2 & \cdot & \cdot & \sum x_k^2 \end{bmatrix}$$

2 – Estimation of Parameters $\hat{B} = (X^t X)^{-1} (X^t Y)$

3 – Fitted Values $\hat{y} = X\hat{\beta}$

4 – Error Terms $e = y - \hat{y}$

5 – Sum of Squares of Errors $SSE = e^t e$

6 – Sum of Squares of Regression $SSR = \hat{B}^t X^t Y - n\bar{Y}^2$

7 – Sum of Squares Total $SST = Y^t Y - n\bar{Y}^2$

8 – Coefficient of Determination $R^2 = \frac{SSR}{SST}$ or $R^2 = 1 - \frac{SSE}{SST}$

9 – Variance – Covariance Matrix $C = \sigma_e^2 (X^t X)^{-1}$ where $\sigma_e^2 = \frac{SSE}{n - p}$

10 – Correlation Matrix

$$C.M = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} = \begin{bmatrix} 1 & r_{12} & r_{13} \\ r_{21} & 1 & r_{23} \\ r_{31} & r_{32} & 1 \end{bmatrix}$$

Note that $r_{11} = r_{22} = r_{33} = 1$ and $r_{12} = r_{21}$, $r_{13} = r_{31}$, $r_{23} = r_{32}$

11 – Testing of Parameters of MLRM

Test Statistic $t = \frac{\hat{\beta}_j - \beta_j}{se(\hat{\beta}_j)}$ with d.f = n – p

The $(1 - \alpha)$ 100% **confidence interval** for $\hat{\beta}_j$ is

$$\hat{\beta}_j - t_{\frac{\alpha}{2}(n-p)} \times se(\hat{\beta}_j) \leq \beta_j \leq \hat{\beta}_j + t_{\frac{\alpha}{2}(n-p)} \times se(\hat{\beta}_j)$$

ANOVA FOR REGRESSION ANALYSIS

ANOVA TABLE

S.O.V	D.F	SS	MS	F-Test
Due to Reg.	$p - 1$	SSR	$MSR = RSS/p - 1$	$F = MSR/MSE$
Due to Error	$n - p$	SSE	$MSE = ESS/n - p$	
Total	$n - 1$	SST	$MST = TSS/n - 1$	

Note that

$$SSR = \text{Sum of Squares Regression} = \hat{B}^t X^t Y - n\bar{Y}^2$$

$$SSE = \text{Sum of Squares Errors} = e^t e$$

$$SST = \text{Sum of Squares Total} = Y^t Y - n\bar{Y}^2$$

FREQUENTLY USED CRITICAL VALUES OF Z

Significance Level (α)	Confidence Level ($1 - \alpha$)	One-Tailed Test	Two-Tailed Test
$\alpha = 0.01$	99%	$Z_\alpha = 2.33$	$Z_{\alpha/2} = 2.58$
$\alpha = 0.02$	98%	$Z_\alpha = 2.05$	$Z_{\alpha/2} = 2.33$
$\alpha = 0.05$	95%	$Z_\alpha = 1.65$	$Z_{\alpha/2} = 1.96$
$\alpha = 0.09$	91%	$Z_\alpha = 1.34$	$Z_{\alpha/2} = 1.70$
$\alpha = 0.10$	90%	$Z_\alpha = 1.28$	$Z_{\alpha/2} = 1.65$

COMMON PROBABILITY DISTRIBUTIONS

Probability Distribution	p.m.f or p.d.f	Mean	Variance
Uniform (discrete)	$P(X = x) = 1/N, \quad N = 1, 2, \dots$	$E(X) = (N + 1)/2$	$V(X) = (N^2 - 1)/12$
Bernoulli	$P(X = x) = p^x q^{1-x}, \quad x = 0, 1$	$E(X) = p$	$V(X) = pq$
Binomial	$P(X = x) = {}^n C_x p^x q^{n-x}, \quad x = 0, 1, 2, \dots, n$	$E(X) = np$	$V(X) = npq$
Poisson	$P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!}, \quad x = 0, 1, 2, \dots, \infty$	$E(X) = \lambda$	$V(X) = \lambda$
Geometric	$P(X = x) = p q^{x-1}, \quad x = 1, 2, 3, \dots, \infty$	$E(X) = 1/p$	$V(X) = q/p^2$
Uniform (continuous)	$f(x) = 1/b - a, \quad a \leq x \leq b$	$E(X) = (a + b)/2$	$V(X) = (a - b)^2/12$
Exponential	$f(x) = \lambda e^{-\lambda x}, \quad x \geq 0$	$E(X) = 1/\lambda$	$V(X) = 1/\lambda^2$
Gamma	$f(x) = \frac{\lambda^p}{\Gamma(p)} x^{p-1} e^{-\lambda x}, \quad x > 0$	$E(X) = p/\lambda$	$V(X) = p/\lambda^2$
Beta	$f(x) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1} (1-x)^{\beta-1}, \quad 0 < x < 1$	$E(X) = \alpha/\alpha + \beta$	$V(X) = \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}$
Normal	$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}, \quad -\infty < x < \infty$	$E(X) = \mu$	$V(X) = \sigma^2$